1. (a) A(5) $\frac{\mu m}{r^2} = m r \omega^2$

$\Rightarrow \omega = \sqrt{\frac{\mu}{r^3}} = \sqrt{\frac{4\times10^5}{7{,}000^3}}$

$\Rightarrow$ (1) $\omega = 1.0799\times10^{-3}$ rad/sec

(2) $V = r\omega = 7{,}000 \times 1.0799\times10^{-3} = \boxed{7.5593 \text{ km/sec}}$

(2) $T = \frac{2\pi}{\omega} = \boxed{5.8183\times10^3 \text{ sec} = 9.6972\times10 \text{ min} = 1.6162 \text{ hr}}$

B(15) Let $(r=R,\ v=\sqrt{\frac{\mu}{R^3}}t,\ \lambda=0)$ be an operating point

and $\begin{cases} r = R+\Delta r \\ v = \sqrt{\frac{\mu}{R^3}}t + \Delta v \\ \lambda = 0+\Delta\lambda \end{cases}$ ①

Introducing ① into EOM leads to

i) $(\ddot{R}+\Delta\ddot{r}) - (R+\Delta r)\left(\sqrt{\frac{\mu}{R^3}}+\Delta\dot{v}\right)^2 \underbrace{\cos^2\Delta\lambda}_{\cong 1} - (R+\Delta r)(\Delta\dot{\lambda})^2$

$= -\frac{\mu}{(R+\Delta r)^2} + \frac{F_r}{m}$

$\Rightarrow \Delta\ddot{r} - R\left(2\sqrt{\frac{\mu}{R^3}}\Delta\dot{v}\right) - \Delta r\frac{\mu}{R^3} - \cancel{\frac{\mu}{R^2}} = -\cancel{\frac{\mu}{R^2}} + \frac{2\mu}{R^3}\Delta r + \frac{F_r}{m}$

(4) $\Rightarrow \Delta\ddot{r} - 2\sqrt{\frac{\mu}{R^3}}R\Delta\dot{v} - 3\frac{\mu}{R^3}\Delta r = \frac{F_r}{m}$ 2 - ①

ii) $(R+\Delta r)\Delta\ddot{v}\cos\Delta\lambda + 2(\dot{R}+\Delta\dot{r})\left(\sqrt{\frac{\mu}{R^3}}+\Delta\dot{v}\right)\cos\Delta\lambda$

$-2(R+\Delta r)\left(\sqrt{\frac{\mu}{R^3}}+\Delta\dot{v}\right)\Delta\dot{\lambda}\underbrace{\sin\Delta\lambda}_{\simeq\Delta\lambda} = \frac{F_v}{m}$

(4) $\Rightarrow R\Delta\ddot{v} + 2\sqrt{\frac{\mu}{R^3}}\Delta\dot{r} = \frac{F_v}{m}$ 2 - ②

iii) $(R+\Delta r)\Delta\ddot{\lambda} + 2(\dot{R}+\Delta\dot{r})(\Delta\dot{\lambda}) + (R+\Delta r)\left(\sqrt{\frac{\mu}{R^3}}+\Delta\dot{v}\right)^2 \underbrace{\sin\Delta\lambda\cos\Delta\lambda}_{\simeq\Delta\lambda} = \frac{F_\lambda}{m}$

(4) $\Rightarrow R\Delta\ddot{\lambda} + \left(\frac{\mu}{R^3}\right)R\Delta\lambda = \frac{F_\lambda}{m}$ 2 - ③

Introducing $\omega = \sqrt{\frac{\mu}{r^3}}$ into 2 −1, 2, 3 leads to

In-Plane Dynamics

$$\Delta\ddot{r} - 2\omega R\Delta\dot{v} - 3\omega^2 \Delta r = \frac{F_r}{m}$$

$$R\Delta\ddot{v} + 2\omega\Delta\dot{r} = \frac{F_v}{m}$$

Out-of-Plane Dynamics

$$R\Delta\ddot{\lambda} + \omega^2 R\Delta\lambda = \frac{F_\lambda}{m}$$

Define $(x \equiv \Delta r,\ y = R\Delta v,\ z = R\Delta\lambda)$

Then,

$$\left\{\begin{aligned} \ddot{x} - 2\omega\dot{y} - 3\omega^2 x &= \frac{F_r}{m} \\ \ddot{y} + 2\omega\dot{x} &= \frac{F_v}{m} \\ \ddot{z} + \omega^2 z &= \frac{F_\lambda}{m} \end{aligned}\right.$$

2 (40) $T = 24\,hrs = 86{,}400\,sec$

(5) $\omega = \frac{2\pi}{T} = 7.2722\times10^{-5}$ rad/sec

$$\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} = \begin{bmatrix} -110 \\ 70 \\ 10 \end{bmatrix} km \quad \begin{bmatrix} \dot{x}_0 \\ \dot{y}_0 \\ \dot{z}_0 \end{bmatrix} = \begin{bmatrix} -10 \\ 20 \\ 1 \end{bmatrix} m/sec = \begin{bmatrix} -0.01 \\ 0.02 \\ 0.001 \end{bmatrix} km/sec$$

(5) $\dot{y}_0^+ = \cdots = -3.7243\times10^{-4}$ km/sec

$\dot{x}_0^+ = \cdots = 0.0226$ km/sec

$\dot{z}_0^+ = \cdots = -0.0013$ km/sec

Check!

$x(x_0, y_0, z_0, \dot{x}_0^+, \dot{y}_0^+, \dot{z}_0^+, t = 2hr = 7{,}200sec) = 0$

$y(\quad \text{"} \quad) = 0$

$z(\quad \text{"} \quad) = 0$

$\Delta\vec{V}_0 = (\dot{x}_0^+ - \dot{x}_0)i + (\dot{y}_0^+ - \dot{y}_0)j + (\dot{z}_0^+ - \dot{z}_0)k$

(10) $\Rightarrow \underline{\Delta\vec{V}_0 = 0.0326i - 0.0204j - 0.0023k}$ km/sec

Take a derivative of HCW-eqn.

(5) $\Rightarrow \begin{cases} \dot{x}(t) = \dot{x}_0 \cos\omega t + (3x_0\omega + 2\dot{y}_0)\sin\omega t \\ \dot{y}(t) = -2\dot{x}_0 \sin\omega t + (6\omega x_0 + 4\dot{y}_0)\cos\omega t - (6\omega x_0 + 3\dot{y}_0) \\ \dot{z}(t) = -z_0\omega \sin\omega t + \dot{z}_0 \cos\omega t \end{cases}$

(5) $\begin{cases} \dot{x}(t_f^-; x_0, y_0, z_0, \dot{x}_0^+, \dot{y}_0^+, \dot{z}_0^+) = \dot{x}_f^- = 0.0072\,km/sec \\ \dot{y}(t_f^-; \quad \text{"} \quad) = \dot{y}_f^- = -0.0164\,km/sec \\ \dot{z}(t_f^-; \quad \text{"} \quad) = \dot{z}_f^- = -0.0015\,km/sec \end{cases}$

$\Rightarrow \Delta\vec{V}_f = (0 - \dot{x}_f^-)i + (0 - \dot{y}_f^-)j + (0 - \dot{z}_f^-)k$

(10) $\Rightarrow \underline{\Delta V_f = -0.0072i + 0.0164j + 0.0015k \quad km/sec}$

<~~Spacecraft Dynamics and Control~~>

~~Final Examination~~

4

3. (20)

A (15)

(5) The S/C is not in equilibrium, since the midway point between the Earth and Moon is not one of five Lagrangian point.

The position and velocity ~~vector~~ of the S/C is given as

$$(x, y, z) = (\mu - 0.5, 0, 0)$$

$$(\dot{x}, \dot{y}, \dot{z}) = (0, 0, 0)$$

Introducing these conditions into the dynamics, we have

$$\ddot{x} = (\mu - 0.5) + \frac{0.5(1-\mu)}{r_1^3} - \frac{0.5\mu}{r_2^3}$$

$$\ddot{y} = 0$$

$$\ddot{z} = 0$$

$$r_1 = 0.5, \quad r_2 = 0.5$$

Therefore, the acceleration vector of the S/C is

$$\ddot{\vec{r}} = \left[(\mu - 0.5) + \frac{(1-\mu)}{0.5^2} - \frac{\mu}{0.5^2}\right] i$$

$$= \left[(\mu - 0.5) + \frac{1-2\mu}{0.5^2}\right] i$$

(10) $\Rightarrow$ $\ddot{\vec{r}} = 3.4139\, i$

B. (5) In order for the S/C to be in equilibrium, we need to cancel the natural acceleration $\ddot{\vec{r}} = 3.4139\, i$

$\therefore \quad \ddot{\vec{r}}_{required} = -3.4139\, i$

4 (20) The solution to

$$\Delta\dot{x} = A\Delta x, \quad \Delta x(0) = \Delta x_0$$

Can be expanded in eigenspace as

$$\Delta x(t) = e^{At}\Delta x_0$$
$$= a_1 \xi e^{j\omega t} + a_2 \bar{\xi} e^{-j\omega t} \quad (a_1, a_2) \in R$$
$$\Rightarrow \Delta x(t) = a_1 \xi (\cos\omega t + j\sin\omega t) + a_2 \bar{\xi} (\cos\omega t - j\sin\omega t)$$
$$= (a_1\xi + a_2\bar{\xi})\cos\omega t + j(a_1\xi - a_2\bar{\xi})\sin\omega t$$

(5)

$\Rightarrow$

$$\Delta x(0) = a_1\xi + a_2\bar{\xi}$$
$$\Delta\dot{x}(0) = j\omega(a_1\xi - a_2\bar{\xi})$$

(5)

$\Rightarrow$ Can solve for (a_1, a_2)

If $\begin{cases} a_1 = a_2 & \Rightarrow \Delta x(0) = 2a_1 \mathrm{Re}\,\xi \\ a_1 = -a_2 & \Rightarrow \Delta x(0) = 2a_1 \mathrm{Im}\,\xi \end{cases}$

$\Rightarrow$ $x(t)$ is periodic with frequency ω.

Ultimately If $\Delta x(0) \in \mathrm{span}\{\mathrm{Re}\,\xi, \mathrm{Im}\,\xi\}$, then the solution is periodic with frequency ω. (10)